

Your Mileage May Vary: The Positives and Negatives of Going Electric

BEVs today are better than they've ever been, but they still need to compete with conventional vehicles. What attracts consumers to purchase a BEV over an ICEV or even a hybrid car? Is it the hype? Is it the appeal of shiny, new technology? Is it because we feel good about changing the world with the purchases we make?

The BEV checks all of those boxes. For so many, it's the obvious path forward to a sustainable, carbon-free transportation future. But there are pros and cons to all technologies, and there are unintended consequences to pushing any technology too far, especially if it's not yet ready. In this chapter, we will investigate the various pros and cons of battery electric vehicles and take a look at the BEV market both today and in the future.

The Electric Advantage

Let's start with the advantages. This list is not meant to be exhaustive, but to highlight some of the main benefits that electric vehicles can offer in the light-duty sector. You will notice that with many of the advantages (and disadvantages later on) we follow with a "but ..." This is not meant to take away from these benefits—rather, it's an attempt to provide a complete picture on where the battery electric vehicle stands today.

No Exhaust Emissions

As we all now know, a sure-fire way to determine if a car is fully (battery) electric is to check for a tailpipe. No tailpipe? Then you can be certain it's a BEV (we may have just dated this book; who knows what other non-tailpipe technologies may come in the future!). And although there are many other factors that contribute to the total emissions of a vehicle, the lack of tailpipe means BEVs have no *exhaust* emissions. This can result in cleaner air locally and, depending on how the battery is produced, how the electricity used to charge the car is generated, etc., a lower overall carbon footprint.

Reduced Noise

If you drive a conventional vehicle, you know it makes noise. This noise comes from three main sources:

1. The engine
2. Wind
3. The tires and the road

At higher speeds, wind resistance and tire noise make up a good portion of the overall sound of a vehicle. However, at lower speeds (and when idle), it's all about the rumble of the engine. Electric cars will experience similar wind and tire noise, but because they don't have a combustion engine, they are quiet at low speeds! Imagine a city center with little to no vehicle noise. In practice this is actually both an advantage and a disadvantage. Reduction in noise is typically considered a good thing, and noise itself is regarded as "pollution" at high levels. But, this comes with an unintended consequence. Researchers have reported that silent BEVs are 40% more likely than other vehicles to be involved in accidents with pedestrians (particularly individuals with vision impairments) [1]. As a result, auto companies are adding artificial noise at low speeds for safety purposes. Since July 2019, this has been required by law in the EU.

Renewable Energy

The electricity used to charge your plug-in can be generated in many different ways, from either renewable or non-renewable sources. Globally, the majority of electricity (about 63% [2]) is made from fossil fuels, mainly coal and natural gas, which are non-renewable resources. And if that's not bad enough, burning these fuels results in anthropogenic carbon emissions. That's a fancy way of saying that human activity is adding carbon into the atmosphere beyond its natural flux.

Renewable electricity generation, such as wind, solar, and hydropower, is sustainable (at least we have no reason to believe that the wind will stop blowing, the sun will stop shining, or water will stop flowing—not permanently anyway) and does not *directly* produce anthropogenic carbon emissions. Thus, the more renewable electricity we use to charge our electric vehicles, the less carbon we put into the atmosphere.

Conventional fossil-based gasoline and diesel fuel can never be renewable, but some electricity already is today.

While we have this listed as an advantage of electric vehicles, it's a bit more complicated than that. For example, wind and solar are intermittent resources, because, on a day-to-day basis, the wind isn't always blowing, nor is the sun always shining. To rely on a growing capacity of these renewables, large-scale storage systems will likely be required for storing excess energy and releasing it on demand. How can we store this energy? One option is batteries! This is another way that improved battery technology can help reduce transportation carbon emissions.

Looking at hydropower, a surprising (at least to us) unintended consequence comes from a hidden source of greenhouse gases. Recent studies (e.g., [3]) have shown that man-made reservoirs can lead to tiny microorganisms breaking down organic matter (such as trees and grass) in the water in the absence of oxygen, leading to methane as a byproduct (recall that methane is a much more potent GHG than CO₂). All in all, the study indicates that 1.3% of all man-made GHGs come from this source. A small, but not trivial, amount! Of course, others argue that the science of reservoir emissions is far from settled, and some research suggests that reservoirs can even act as carbon sinks, absorbing carbon from the atmosphere [4].

This is not to say that we shouldn't be pushing for more renewable electricity. We have this in the advantages column for a reason! However, when we compare the impact of electric vehicles and their electricity origins, all sources of GHGs must be considered to make the best possible decisions going forward.

Efficiency

If we look at just the car, BEVs are significantly more efficient than ICEVs. BEVs convert about 70–85% of the electrical energy from the grid to power the wheels (accounting for charging losses) [5]. A conventional ICEV converts about 12–35% of the energy stored in gasoline to power the wheels (a diesel car converts up to 45%). This increases to 21–40% for a gasoline full hybrid [6]. Note that both BEVs and hybrids benefit from regenerative braking, which recovers some of the energy that would normally be lost. We'll do a more detailed comparison of the efficiency of the different propulsion types in Chapter 11 at both the vehicle and system levels. But for our purposes here, BEVs are the clear winner for vehicle efficiency.

Fuel Cost

Electricity is cheaper than gasoline. At the time of writing, it costs about half as much to fuel an electric vehicle as it does a conventional vehicle in the USA [7].

It's important to note that a large portion of the price you pay for a gallon of gasoline goes to taxes. In the great state of Wisconsin (in September 2020), a gallon of regular gasoline costs \$2.07. About 33 cents of that goes to state tax, and about 18 cents goes to federal tax, for a total of around 51 cents per gallon. That's 25% of the total cost! What do these taxes go toward? They're partially used to fund road repair and construction. This begs the question: What will happen to our roads if more and more people stop paying for gasoline because they've switched to BEVs?

Outside of the USA, the situation varies, but in the UK, the cost (in September 2020) of a liter of petrol is £1.15. Based on September 2020 currency conversions, this is the equivalent of paying \$4.26 for a (US) gallon of regular gasoline. In the UK, taxation is approximately 70% of the price, so about \$3.00 of tax per gallon. This tax is not allocated to roads or anything specific; rather it is part of the total tax take for general expenditure. This fuel tax amounted to £28.4 billion (\$36.4 billion), or 3.5% of the total tax take, in 2019–2020.

While BEVs are significantly cheaper to fuel today, as they grow in numbers, governments will need to start recuperating the lost public funds somehow. It seems like a safe bet that this will increase BEV operational costs at some point in the future. But for the time being, lower fuel cost is an advantage of the battery electric vehicle.

Less Maintenance

BEVs require less routine maintenance than conventional, or even hybrid, vehicles. There are fewer moving parts in a BEV than there are in a vehicle with an ICE. There are also fewer fluids to change. That oil change that your car is currently past due for? That's not an issue for BEVs. If your battery uses a liquid coolant, however, you may need to regularly check that. Brakes may also require less maintenance as regenerative braking used in BEVs reduces brake wear.

On the other hand, as we learned in Chapter 4, heavier vehicles produce more non-exhaust particulates from, among others sources, tire wear. While we haven't seen any data yet that shows BEVs require more frequent tire changes, deductive reasoning would suggest it.

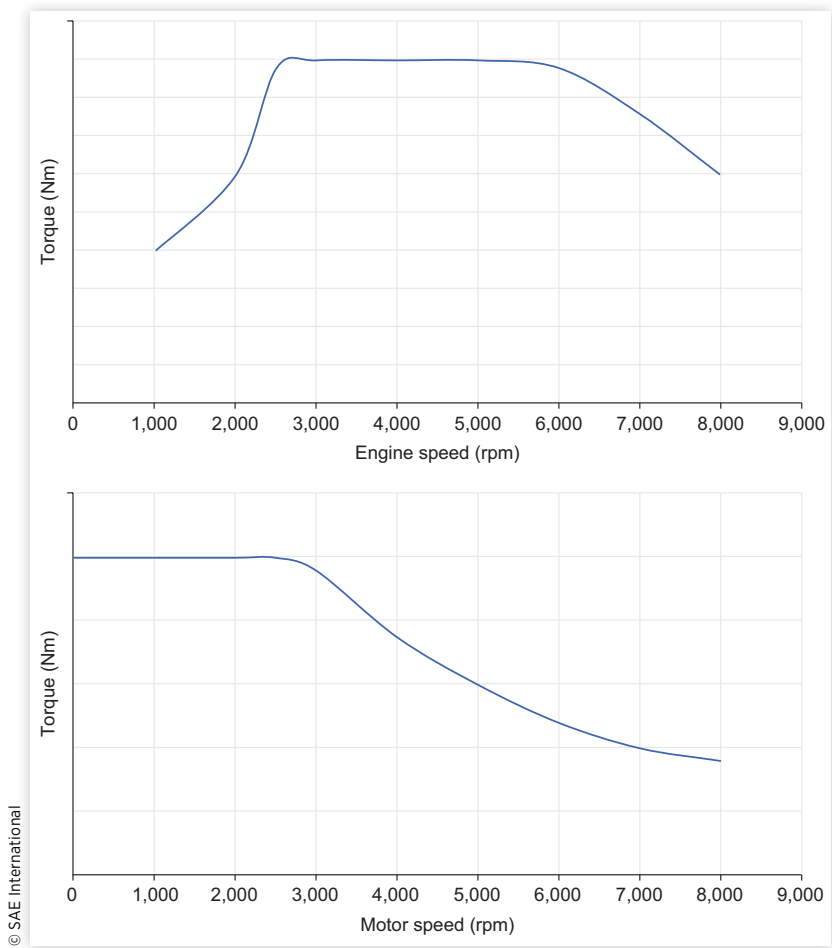
The battery itself may need to be replaced at some point in the car's lifetime, because a battery can only be charged and discharged so many times before it reaches the end of its life. There is much debate about how long a BEV battery pack will last, so in the absence of data (there still are not that many BEVs on the road in 2020 and even fewer ten-year-old BEVs), we use the warranty period as a good estimate. For example, Tesla offers warranties of eight years and 100-150k miles (depending on the model) on its batteries. Note that this warranty only guarantees 70% battery capacity over this period [8]. As we will learn in Chapter 11, whether a battery needs to be replaced over a BEV's lifetime can make a big difference in its total carbon footprint (as well as cost to the consumer!). But more on that later.

Acceleration

If your 0–60 time is important to you, electric cars usually have the upper hand, because a BEV generates instant torque ([Figure 7.1](#) compares the torque curves of ICEVs and BEVs). According to the website FastestLaps, at the time of writing the Aspark Owl (shown in [Figure 7.2](#)), a Japanese-made BEV, holds the quickest 0–60 time of 1.7 seconds [9]. Of course, the Owl is a hypercar¹—Aspark is only selling about 50 of them for around \$3.2 million each. Coming back to the real world (well, the

¹ A hypercar usually costs more than one million US dollars, has a top speed of over 200 mph (likely well over), and is typically produced in a quantity of less than a couple hundred.

FIGURE 7.1 Comparison of representative torque curves for an ICEV (top) and a BEV (bottom).



real world if you have over \$90,000 to spend on a new car), MotorTrend measured the 0–60 times for a number of production cars in 2019, and a Tesla Model S Performance with Ludicrous mode out-accelerated all others, clocking in at 2.4 seconds (Tesla currently has the Model S Performance 0–60 time listed as 2.3 seconds on their website) [10].

This doesn’t mean that BEVs are faster than their conventional counterparts, and, in fact, ICE-powered vehicles will typically win a race of any significant distance. Currently, the fastest road-legal production car (of which at least 20 have been produced) is the Bugatti Chiron Super Sport, powered by a 16-cylinder engine and achieving a top speed of nearly 305 mph [11]. For comparison, the Aspark Owl has a top speed of 249 mph. The Rimac C_Two (also shown in Figure 7.2), which has the highest top speed of any BEV, can reach 258 mph [12].

FIGURE 7.2 An Aspark Owl (left) and a Rimac C_Two (right).

Fill Up at Home

If you have the right setup and a commute that isn't too crazy, you may be able to get by with only charging your electric car at home (just don't forget to plug it in every night!). For those who find it annoying to have to periodically stop to fill up at the gas station, this could be an advantage.

The Electric Disadvantage

In addition to the advantages we just described, there are several downsides to owning an electric car today. Again, this is not an exhaustive list. Interested readers should do further research on the current disadvantages (and advantages) of BEVs. Of course, some, many, or even all of these disadvantages may diminish in the future as technology improves.

Vehicle Cost

Today, electric cars cost significantly more than conventional vehicles or hybrids. Of course, many models are in the luxury segment without an analogous ICE model (Tesla Model S, Porsche Taycan, Mercedes-Benz EQ, Lucid Air EV, etc.), which makes it difficult to directly compare cost. However, there are a few BEVs today with a conventional or hybrid counterpart. For example, the 2020 Kia Niro (an FHEV) has a starting MSRP of \$24,590. The 2020 Kia Niro BEV has a starting MSRP of \$39,090. That's a 60% increase in cost for the BEV!

The 2021 Hyundai Kona starts at \$20,500, while the 2021 Kona Electric starts at \$37,390. In addition, the 2020 Hyundai Ioniq Hybrid starts at \$23,200, while the 2020 Ioniq Electric starts at \$33,045. Most people cannot justify this significantly higher up-front cost,² even if the fuel cost is lower for the BEV. However, it is likely that the gap in price between car types will shrink as technology continues to evolve. Battery prices have fallen significantly in the last decade, as was shown in Figure 6.10, and it is expected that they will continue to fall.

² Note that today, government subsidies can significantly reduce the cost of certain BEVs to the consumer, as we'll discuss in Chapter 12.

Range Anxiety

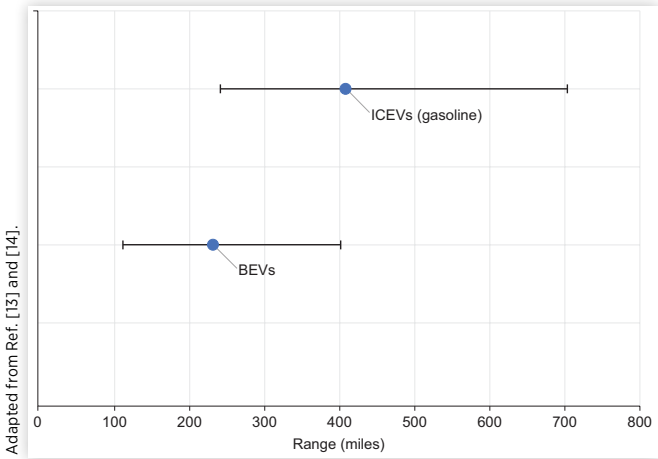
As previously mentioned, the range of BEVs has substantially increased in the last decade. Many would even argue that the BEVs of today satisfy the majority of trip distances. However, as shown in [Figure 7.3](#), the median range of all-electric vehicles is still significantly lower than conventional (ICEV) vehicles.

Given their lower range, multicar households seem more inclined to purchase a BEV as a second or even third vehicle. This is more of an anecdotal observation, as data on this particular topic are hard to come by, but our perception is echoed in a 2017 study by Karlsson [15]. That work highlights the importance of multicar households for BEV adoption beyond early adopters. The combination of low range, high charge times, and lack of infrastructure, can make it difficult to rely solely on a BEV for all of your driving needs.

An interesting concept that could help with range anxiety is “battery swapping.” This allows for an immediate swap between a discharged battery and a charged one. Imagine if you could pull into a swapping station and within just a few minutes have a fully charged battery ready to go without leaving your car! No more waiting 30 minutes or even hours for charging.

Battery swapping has been around in various incarnations for nearly as long as BEVs themselves (for example, the Hartford Electric Light Company was swapping batteries way back in 1896). The first modern company to deploy this concept commercially was called Better Place, which was founded in 2007 in California but operated in Israel. Their offering was coupled with the Renault Fluence ZE, the only model available for battery swapping at the time. Better Place subscribers paid around \$350 per month to lease access to the batteries and the switching stations [16]. In 2013, after \$850 million in investments, Better Place filed for bankruptcy. There seem to

FIGURE 7.3 Comparison of minimum, maximum, and median ranges for gasoline ICEVs [13] and BEVs [14] for models available in the USA. BEV data are from 2020 models, whereas ICEV data are from 2016 (we don't expect that the ICEV ranges have changed much overall in four years, so this comparison is valid).



be several reasons for the company's downfall, including the massive investment to develop the swapping infrastructure and the low market penetration of compatible cars. As noted in a 2013 article in *The Atlantic*, "you could drive any electric car you wanted as long it was a Renault Fluence ZE" [17].

The Chinese electric car company, Nio, believes that Better Place failed because the BEV market was too small at that time, and also because they didn't make their own cars [18]. Nio has recently released their own spin on battery swapping, called BaaS (battery as a service). Under this model, car buyers can save \$10,000 because (and we are so happy we can finally insert this phrase into our book) "batteries [are] not included." According to Car and Driver, a 70 kWh pack with six swaps per month is priced at around \$142/month [19].

Whether Tesla and other companies figure out a way to get charge times down, or if the world moves to swappable batteries, for now range anxiety continues to be one of the main disadvantages of BEVs.

Charge Times

Speaking of charge times, another significant disadvantage of BEVs is the time required to charge the battery. We're used to filling our gas tanks in 3-5 minutes. Unless you're able to charge at home, waiting for hours to charge your BEV, well, sucks.

How long does it take? There are several variables that affect a BEV's charge time, including the battery size, the starting charge level, the desired (target) charge level, the power of your charging point, and the temperature. To see where we are today, we checked the websites of a few popular BEVs on the market: the Chevrolet Bolt, the Nissan LEAF, and the Tesla Model S.

Let's start with the Chevrolet Bolt. The Bolt has a 66 kWh battery and an EPA-estimated range of 259 miles on a full charge. Chevrolet lists the charge times as [20]:

- Level 1: 4 miles of range per hour
- Level 2: Full charge in approximately 10 hours
- DC fast charge: Up to 100 miles in 30 minutes

The Level 1 option is really just a standard US household outlet with a 120 V connection. For Level 1, you can see that the charge time is listed as the number of miles gained from an hour's worth of charging. This is by far the slowest method—it would take over 60 hours to fully charge your Bolt! This drops to 10 hours for a Level 2 charger, which uses a higher output 240 V power source (standard in many countries). Finally, the DC fast charge option would theoretically complete a full charge in about 1.3 hours. However, many rapid chargers reduce power well before 100% charge for battery protection. This is why you will often see charge times based on an 80% charge.

Why is DC fast charging so, well, fast? Unlike Level 1 and 2 chargers, which require conversion of AC to DC power (does this make anyone else think of a certain hard rock band?) onboard the vehicle, DC fast chargers significantly increase charge speed by providing DC power directly.

FIGURE 7.4 Tesla Supercharger station in Brenner, Italy. If you look closely at the bottom of each charger, there is a number followed by a letter. Chargers with the same number share power—try to avoid doubling up for faster charge times!



For a 2020 Nissan LEAF with a 62 kWh battery and up to 226 miles of range, a Level 2 charger gets you a full charge in 11.5 hours. DC fast charge will get you to 80% capacity in 45–60 minutes, depending on charge power [21].

Finally, checking out the Tesla website, you can charge a Model S up to 80% in about 30 minutes at a Supercharger (their version of DC fast charging—see [Figure 7.4](#)) and up to 44 miles of range per hour at home with their Wall Connector charging system [22]. Supercharger charge times can increase if other cars are using the charge post at the same time. Tesla recommends parking at a Supercharger shared with a car that is almost done charging (or, for that matter, not sharing!). On the other hand, Urban Superchargers (a more compact version designed for highly populated city centers) don't share power, so charge time should be consistent regardless of whether or not someone is charging in an adjoining stall [23].

Infrastructure

As we mentioned earlier, one advantage of BEVs is that you can charge from home. This automatically adds many charging points to the electric vehicle infrastructure. However, as we just saw, it can take a very long time to charge at home with a standard Level 1 outlet. Upgrading to a Level 2 system significantly reduces charge times, but, at the time of writing, you will still have to wait at least 10–12 hours to charge a Bolt or a LEAF.

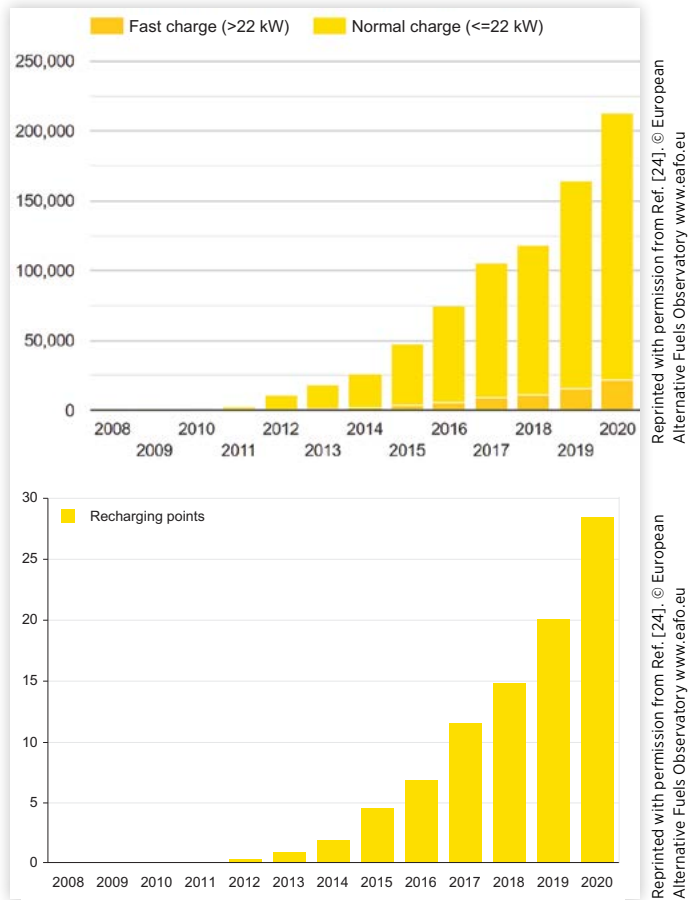
Of course, home charging assumes you have easy access to electricity where your car is parked. What do you do if you live in an apartment building with street parking (you may have seen pictures of apartment dwellers hanging extension cords out of

two-story windows to juice up their cars on the street below!)? Clearly, a public infrastructure is needed if BEVs are going to make up a significant portion of the fleet.

The public charging infrastructure is definitely growing. [Figure 7.5](#) (left-hand side) shows the number of charging outlets in the EU between 2008 and 2020. Both normal chargers (less than 22 kW charge power) and fast chargers (greater than 22 kW charge power) are shown. In just the last five years (since 2015), the number of charging outlets has roughly quadrupled. The data are shown a bit differently on the right-hand side of [Figure 7.5](#). Here, we have the number of charging outlets per 100 km of highway in the EU. In 2020, about 28 cars on average could publicly charge at the same time within 100 km of highway.

In the USA, there are roughly 25,000 public Level 2 charging stations, with more than 75,000 outlets at the time of writing. There are currently about 4,000 public DC fast charging stations, with more than 16,000 outlets [25]. Globally there are more

FIGURE 7.5 Number of normal and fast charging outlets in the European Union (top). Number of fast charging outlets per 100 km of highway in the EU (bottom). Note that outlets are referred to as “points” in the plots [24].



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than 2,000 Tesla Supercharger stations with over 20,000 chargers as of November 2020 [26].

How do these numbers compare to conventional gas stations? There are around 115,000 gas stations in the USA [27]. Let's assume an average of eight gas pumps per station. This puts us at just under one million cars that theoretically could all gas up at the same time, versus around 90,000 Level 2 and DC fast public charge connectors. The electric infrastructure is not there yet, but it's growing rapidly.

Mining

In 2016, Amnesty International raised concerns over the unethical mining of cobalt in their report, "This is what we die for: human rights abuses in the Democratic Republic of the Congo power the global trade in cobalt" [28]. Cobalt is a key ingredient used in the cathodes of lithium-ion batteries due to its thermal stability and high energy density. The report exposed serious human rights issues, including child labor and lack of protective equipment for miners.

Amnesty's report brought much needed attention to a very serious ethical problem. In a subsequent publication, the group reported that the situation had improved somewhat, but there was still much work to be done [29]. Indeed, a few years after those reports, these human rights concerns remain a crucial problem.

In a comprehensive assessment of BEVs versus ICEVs, Arthur D. Little states that, "While the BEV driver reduces their local contribution to greenhouse gas emissions, they create a more diffuse set of environmental impacts spread across the globe, the consequences of which are largely borne by rural and often disadvantaged communities near the mines from which BEV suppliers source raw materials for battery pack manufacturing" [30].

The good news is that the negative environmental and human rights aspects associated with mining activities are now more in focus. Organizations like the World Economic Forum's Global Battery Alliance have proposed steps toward a socially responsible and environmentally sustainable battery industry, including implementing standards across the supply chain and promoting transparency of where and how cobalt is mined and processed [31].

Weather

Extreme weather can significantly impact the range of battery-powered vehicles. Recently, the Norwegian Automobile Federation tested 20 BEVs in Norway's winter conditions. They found that, on average, BEVs lose nearly 20% of their range and take longer to charge in cold weather [32]. In a study by Car and Driver, three tests at 70 mph were performed for a Tesla Model 3 on a 3°C (38°F) day: the first with all HVAC (air conditioning) turned off, the second with HVAC set to 22°C (72°F) on auto, and the third with HVAC on full blast with all five heated seats turned to the highest setting. Compared to the first test, the second test reduced the range by about 15%, while the third test reduced it by about 26% [33]. The company Geotab analyzed 4,200 BEVs and 5.2 million trips and mapped out the impact of temperature on range. They have a neat tool that allows you to pick a car (2019 model or older at the time of

writing) and slide a temperature bar to see the effect. A 2019 Chevrolet Bolt got the most range (261 miles) at an outside temperature of 22°C (71°F). At 42°C (108°F) this dropped to 170 miles. At -22°C (-7°F), it dropped to 113 miles [34].

Another study, by Motoaki et al., quantified the effect of cold weather on charge time for DC fast chargers in the USA [35]. They calculated the charging efficiency for a Nissan LEAF on a median temperature day,³ starting at 20% charged. After a 30-minute charge, the state of charge ranged from as low as 49% in northern states to as high as 74% in southern states, demonstrating a significant deterioration in charging efficiency in cold temperatures.

Weight

Non-exhaust particulate emissions generated from tire, road, and brake wear are only expected to rise as the average weight of cars increases. In a 2020 report titled “Non-exhaust emissions from road transport: causes, consequences and policy responses,” the OECD review several studies that demonstrate the positive relationship between vehicle weight and non-exhaust emissions [36]. BEVs tend to be heavier than their conventional counterparts due to their large battery packs. As a result, a BEV’s non-exhaust particulate emissions are typically higher than an ICEV’s. On the other hand, the regenerative braking used by BEVs tends to reduce brake wear, so this contribution to non-exhaust PM is likely lower compared to an ICEV. Even saving on PM emissions from brakes, though, BEVs still tend to emit greater non-exhaust PM overall than their ICEV counterparts.

Health

In an invited review paper for the British Medical Bulletin, Dr. Sarah Jones asked, “If electric cars are the answer, what was the question?” Jones looked at everything from road traffic injuries to air pollution and concluded, “Overall, the negative health consequences of electric cars seem likely to be at least those of internal combustion engine cars” [37]. We’ll look at the health and environmental concerns surrounding BEVs in depth in Chapter 11.

Today’s Plug-In Vehicle Market

Given their many pros and cons, where do BEVs stand today? If you read the headlines (especially during the COVID-19 pandemic when much of this book was written), you might believe that they have already taken over. Articles show massive growth in BEVs in some countries, while the numbers for conventional vehicles are down. But here’s the trick—by reporting sales in percentages you can make small numbers look big. Two hundred percent growth? Amazing! But unfortunately, an increase of, say, 200% starting from a small number is still a small number. The book *The Art of Statistics: How to Learn from Data* by David Spiegelhalter comes to mind [38].

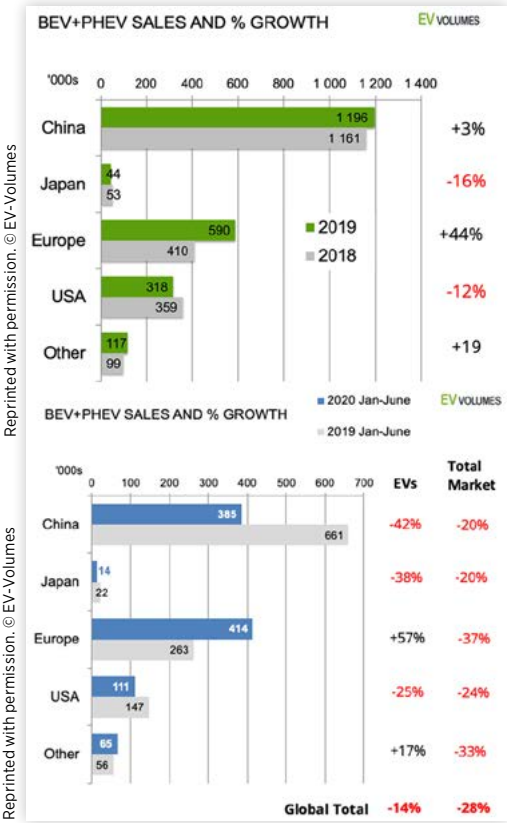
³ We will spare you the mathematical details of how this is determined, but this temperature is the median of the daily maximum hourly temperatures over a year at a specific location. Charging efficiency is expected to be better than this day on half of the days in a year and worse on the other half.

Globally, 2.26 million plug-in vehicles (remember this includes both BEVs and PHEVs) were sold in 2019 [39], up just 9% from 2018. This represents 2.5% of the total light-duty vehicle sales in 2019. Over half of these sales took place in China, highlighting China’s significant role in the developing BEV sector.

The left-hand-side of Figure 7.6 shows the sales and percent growth of plug-ins (2018-2019) for various regions throughout the world. As you can see, Europe had the largest growth in plug-in sales, growing by 44% from 2018 to 2019. China saw minor growth of 3%, while sales actually decreased in Japan and the USA by 16% and 12%, respectively.

It’s interesting that the global increase of plug-ins from the previous year, 2017–2018, was a whopping 64%, which dwarfs the 9% growth from 2018 to 2019. Furthermore, plug-ins had enjoyed a healthy 45–70% growth over the six years prior to 2019 [39]. So what’s with the slowdown? Perhaps we are converging to a steady plug-in market share? Although, if that’s the case, it’s likely that political pressure will result in an increased growth percentage once again. Additionally, Europe in particular does not seem to be slowing down.

FIGURE 7.6 Plug-in vehicle sales growth from 2018 to 2019 (top) and from the first half of 2019 to the first half of 2020 (bottom) [39, 40].

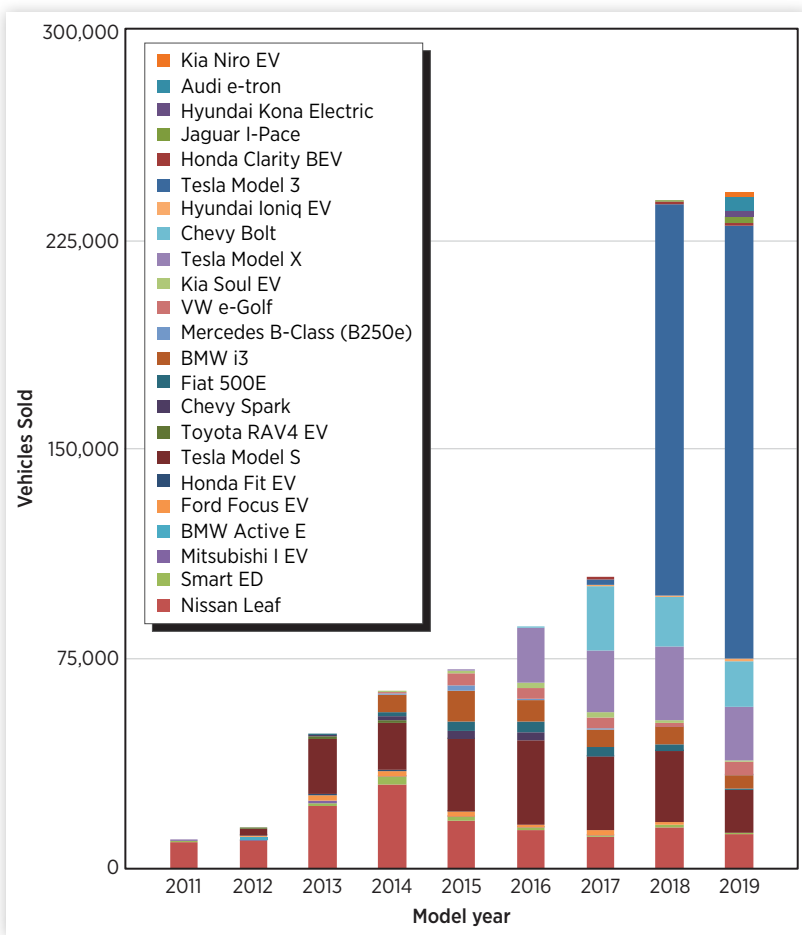


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At the time of writing, we also have data available for the first half of 2020, shown on the right-hand-side of [Figure 7.6](#). We include this only for reference, noting that (1) it's only half-year data and (2) the COVID-19 pandemic has surely influenced the market. Nevertheless, it's interesting to observe that the half-year trends mostly mimic the trends from 2018 to 2019. The exception is China, which saw a 42% decrease in plug-in sales between the first half of 2019 and the first half of 2020.

What BEVs are consumers buying? [Figure 7.7](#) shows the breakdown for the USA. The Nissan LEAF was clearly the market leader between 2011 and 2014, with the Tesla Model S dominating in 2015 and 2016. Deliveries of the Tesla Model X began in late 2015, and by 2017 its sales were similar to both the Chevrolet Bolt and the Model S, with no clear leader in the market—that is, until the Tesla Model 3 was introduced. The Model 3, a smaller, somewhat more affordable version of the Model S, sold more than all other BEV models combined in the USA in both 2018 and 2019. In fact, the Model 3 was the best-selling plug-in car in the world in 2018 [\[41\]](#), with nearly 140,000

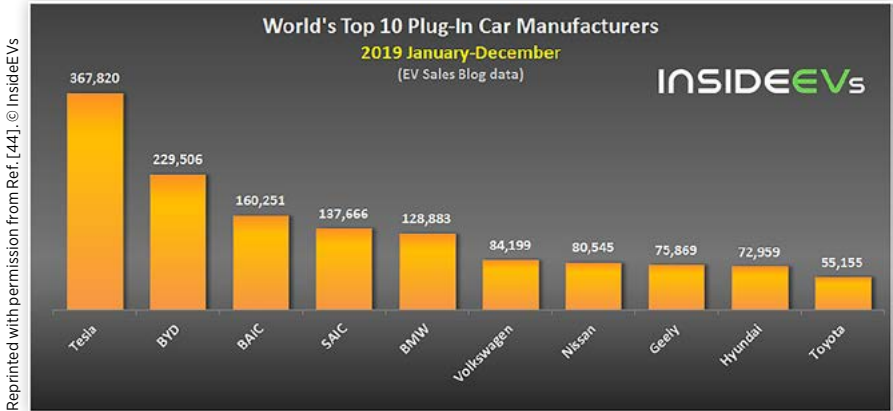
FIGURE 7.7 US BEV sales by model from 2011 to 2019 [\[43\]](#).



Adapted from Ref. [\[43\]](#). U.S. Department of Energy.

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FIGURE 7.8 Global plug-in sales by automaker in 2019 [44].



units sold [42]. Tesla continued to dominate in 2019, selling more plug-in vehicles globally than all other brands, as shown in [Figure 7.8](#), which includes both BEVs and PHEVs in the case where a company produces both. Quite an impressive achievement for a company that makes only BEVs!

If Tesla wins the brand battle, Norway wins the country battle. As shown in [Figure 7.9](#), over 50% of all 2019 new car sales in Norway were plug-ins. BEVs are a good fit for Norway because of their highly renewable electrical grid and relatively short driving distances. The next biggest buyers of plug-ins were Iceland, the Netherlands, and Sweden, with around 18%, 15%, and 11%, respectively. Beyond that, the rest of the countries were lower than 7%.

The impressive growth of BEVs, especially through 2018, has led many to refer to BEVs as a disruptive technology. But how disruptive are they really? Remember, in 2019 they still made up only about 1.7% of all new car sales. Clearly they have yet to take over the transportation landscape! But will they ever? And if so, when? Let's take a look at what's being predicted.

Future Market Projections

From the data we just discussed, it should be clear that ICEs are in the vast majority of vehicles on the road today, as well as in newly sold vehicles. But what does the future hold? If we look back to 2017—the year the media declared combustion dead—then an all-electric fleet should be just around the corner! But even the Economist article [45] that famously declared the death of ICE acknowledged that reaching an all-electric fleet will be a rough ride. While they envisioned driverless electric cars revolutionizing the transportation world, they also conceded that there is much work to be done in buoying up the generating capacity of the world's power grids, implementing new systems of regulations, and dealing with the fallout from disappearing automotive factory jobs. Indeed, most projections show that well over half of the cars on the road in 2040 will still be powered by ICEs. An example is shown in [Figure 7.10](#), which is BloombergNEF's latest projection from their Electric Vehicle Outlook 2020

FIGURE 7.9 Countries with the highest share of new plug-in car sales in 2019.

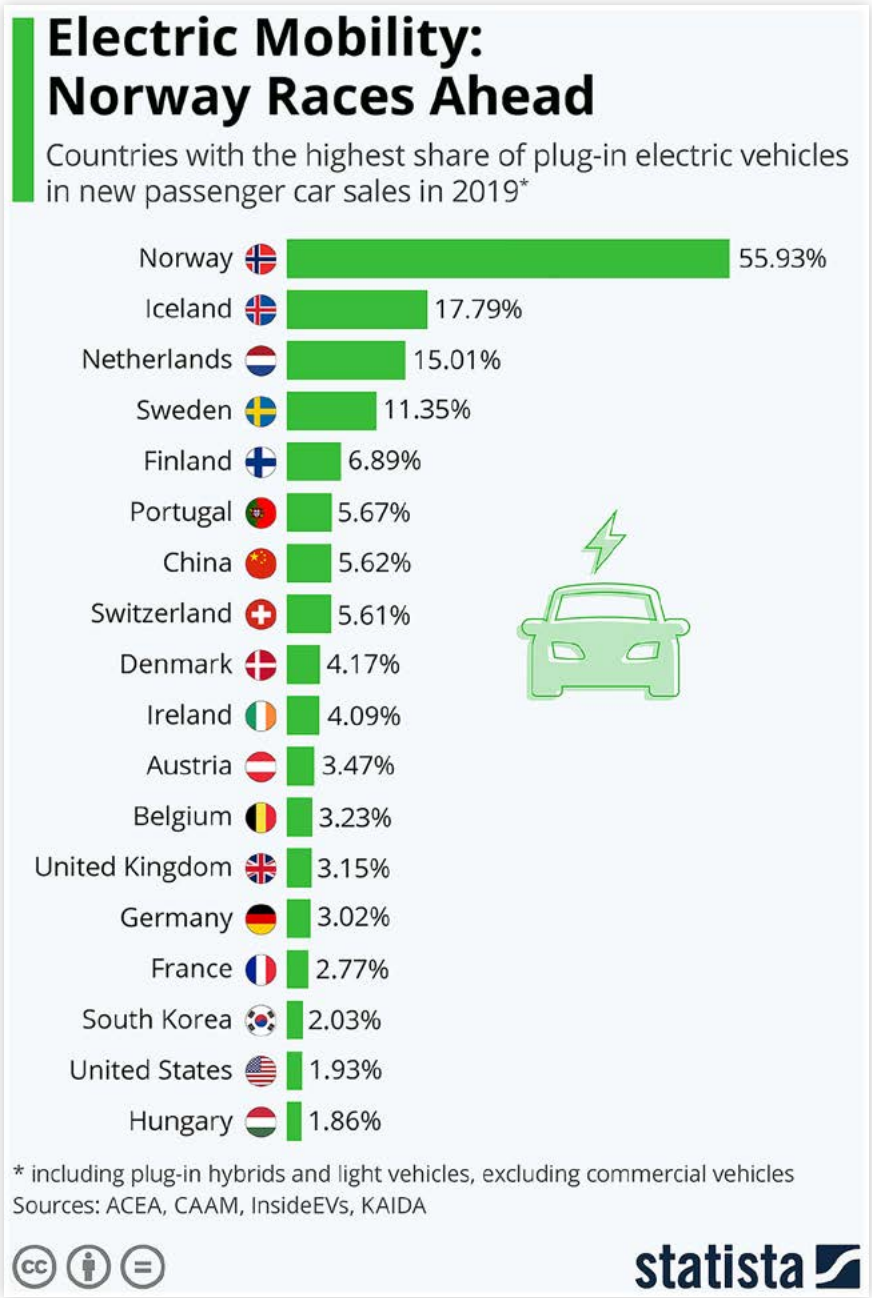
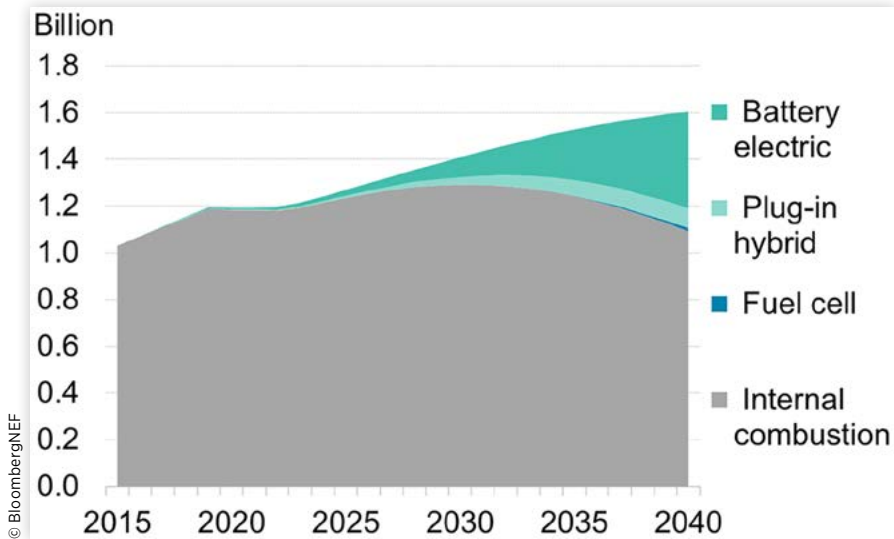


FIGURE 7.10 BloombergNEF projection of the global passenger vehicle fleet by drivetrain [46].



report [46]. Their projections show that, of the 1.6 billion cars on the road in 2040, about 1.2 billion of them are powered by ICE, either in full or as part of a hybrid system.

Beyond projections, there are campaigns and lobbying groups who seek to influence the uptake of plug-ins in the light-duty sector. For example, The EV30@30 Campaign was launched at the 8th Clean Energy Ministerial in 2017 with the aspirational goal of plug-ins making up 30% of new vehicle sales in 2030 within the member countries. Even with this campaign to speed up the deployment of plug-in vehicles, hitting this target would still mean that 70% of new vehicles sold in 2030 would be conventional ICE and full or mild hybrid vehicles (and, of course, the PHEVs will contain an ICE as well). In addition, vehicle quality is going up, so cars are generally lasting longer. The average age of a car on the road is about 12 years old [47]. Even under the EV30@30 scenario, and looking out to 2040, most cars will have ICEs in them. Twelve-year-old cars in 2040 will have been bought in 2028—wouldn't we want those cars to be as clean as possible? Continuing to improve the ICE, alongside battery technology, is essential.

Perhaps the world is moving toward fully electric transportation, but the future is hard to predict. BEVs may or may not live up to their promise. Either way, it appears the transition will take time, and if it never manifests, there needs to be backup plan. Many automakers seem to have the right idea: invest in BEVs, but also continue to invest in alternatives, whether that's hybrids, highly efficient conventional ICEVs, or something entirely different.

Software to Improve BEVs and Beyond

How can we improve BEVs (and all vehicles, really) to help overcome their current pain points? One key area that can help is simulation and modeling (no, not *that* type of modeling—there's a reason Felix and Kelly are engineers!). Advances in modeling

capability, in particular linked to substantial increases in available computing power, have led to huge improvements in all aspects of a vehicle design, including batteries and electric motors. With modeling, we are able to “predict” what will happen using a computational representation of a design. This provides us with a deeper understanding of important processes and also allows us to much more easily analyze a vast array of configurations than is possible (or at least affordable) experimentally. You likely noticed that we put the word *predict* in quotes just above. This is because models are not exact representations of physical systems (some assumptions have to be made in order to solve the equations in any reasonable amount of time). But don’t worry, they’ve evolved to a point where real engineering decisions can be made to support propulsion system design.

There are various ways that modeling is helping elevate not only the performance, but also the safety of electric vehicles. Improving the thermal management of both the battery and the electric motor is an important area where simulation can help. Furthermore, as we mentioned in the previous chapter, under certain conditions a battery pack can enter a thermal runaway situation, leading to vehicle fires. If we simulate this process to determine the risks and develop mitigation strategies, we should be able to significantly improve the safety of these vehicles.

This is where CFD, a specific type of modeling that predicts fluid-flow phenomena based on the laws of fluid motion (think Newton’s second law with forces from gravity, pressure, and viscosity) comes in. It just so happens that Kelly is co-owner of a CFD company called Convergent Science (yes, this is a shameless plug!). One of the company’s main focus areas, at the time of writing, involves predicting thermal runaway using their CONVERGE CFD software [48].

Another key area is systems-level integration. All vehicles are becoming more complex and “smarter” with the abundance of sensors available. A great deal of optimization is required to ensure that these sensors are all used to their full effect, and automakers already expend an enormous amount of effort to ensure this. Vehicle software, as a result of this system integration, is going to become increasingly important. Crucially, the connectivity of future vehicles will enable performance improvements to be implemented remotely, without the user needing to take the vehicle to a garage. Tesla is already leading in this area, with several software updates a year installed remotely on its vehicles. This doesn’t have to be specific to BEVs, either—vehicles of any kind can benefit from this technology, and you can expect to see much more of this in the future.

The Last Mile

In this chapter, we’ve highlighted some of the advantages and disadvantages of battery electric vehicles. We touched on the many attractive features of BEVs and explored where improvements are still needed. We took a look at today’s plug-in market and some of the future market projections. Finally, we very briefly touched on the use of software to help improve performance and safety.

While BEVs have played a starring role in much of the future mobility discussion, many believe that a different solution will disrupt transportation—one based on the most abundant chemical substance in the universe: hydrogen.